

A Corrected Metatheory for Mergeable Replicated Data Types

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Abstract

Mergeable replicated data types reconcile two branch states relative to a common ancestor. The original bottom-up linearization argument for this model does not hold: a global arbitration relation can change when an event outside the version being linearized absorbs an edge. We give a corrected Lean 4 metatheory based on set-relative arbitration, a reachable store invariant, and a contextual ternary Join lemma. We keep the operational MRDT signature free of client invariants and applicability fields. Datatype implementers instead supply issuance, convergence, and an independent sequential specification. Optional safety certificates remain separate. The framework also supplies canonical virtual LCAs and an asynchronous distributed collector for commit history. Datatype-state garbage collection is optional and composes with the history collector by trace erasure. All theorem names in this paper are checked by the repository's `RefactorLedger.lean` build target.

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1 Result and evidence boundary

An MRDT stores immutable versions in a Git-like directed acyclic graph. An update extends one replica head. A merge of branch states a and b reads the state l at their common ancestor and computes $\text{mergeL}(l, a, b)$. The client expects the state at every registered version to equal a legal sequential fold of exactly that version's events.

Figure 1 is the running execution shape. The graph is not decoration: apply creates a one-parent version, merge creates a two-parent version, and a later merge may use an old version as its base. This is why the correctness invariant ranges over the store rather than only the current heads.

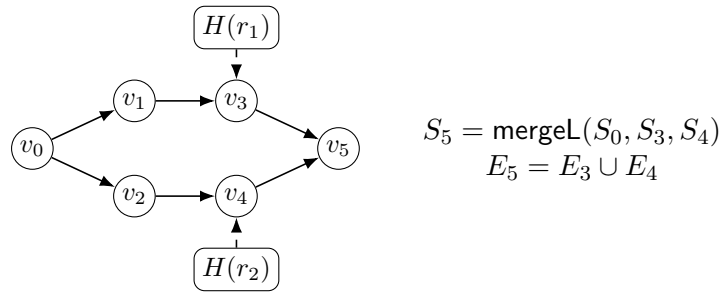


Figure 1: A version-DAG execution. Solid arrows point from parents to children. The merge at v_5 reads the registered LCA v_0 and both branch heads.

The current artifact separates four kinds of evidence.

Layer	Framework supplies	Datatype supplies
Raw execution	version DAG, apply, merge, query	state and operations
Convergence	canonical adequacy, virtual LCA	contextual Join proof
Client protocol	issuance-certified execution	origin issuance relation
Client meaning	certificate interfaces	sequential and safety proofs
Commit GC	distributed protocol and refinement	none
State GC	generic trace-erasure theorem	optional physical protocol

This boundary matters. A theorem about a semantic MRDT does not by itself verify an editor widget, a wire decoder, or crash recovery. The JavaScript runtime is tested against the model-facing fixtures, but it is not extracted from Lean. Its evidence manifest names the exact Lean package associated with each released datatype and labels representation experiments separately. We call Lean declarations *machine-checked*, runtime test results *validated*, and benchmark observations *measured*.

2 Raw MRDT semantics

2.1 The signature stays plain

Definition 2.1 (MRDT signature). *An MRDT signature contains a state type Σ , initial state σ_0 , application operation, query interface, conflict-resolution relation, binary merge inherited from the*

CRDT core, and ternary merge

$$\text{mergeL} : \Sigma \rightarrow \Sigma \rightarrow \Sigma \rightarrow \Sigma.$$

The compatibility law $\text{mergeL}(\sigma_0, a, b) = \text{merge}(a, b)$ relates the ternary and initial-slice binary operations. *MRDTSig*

The signature has no fields named invariant or applicable. A client can invoke the raw update on every state. Generation restrictions enter only in the certified execution of Section 5. This choice prevents the datatype carrier, client protocol, and safety theorem from becoming one circular definition.

2.2 Configurations and transitions

A paper-level configuration is

$$C = \langle V, H, P, \text{vis} \rangle, \quad V(v) = (S_v, E_v),$$

where V is the finite version store, H maps replicas to heads, P maps a version to its parents, and vis is event visibility. Replica state and replica events are derived as $S_{H(r)}$ and $E_{H(r)}$. The Lean **Configuration** also caches these two projections as $\mathbf{N}$ and $\mathbf{L}$; **head_coherent** requires the caches to equal the derivation above. We omit the caches from the rules because they contain no independent semantic information.

Version numbers increase along parent edges. The configuration requires immutable event timestamps, per-replica causal order, coherent heads, and the key LCA equation. If v_T is an LCA of v_1 and v_2 , then

$$\mathcal{E}(v_T) = \mathcal{E}(v_1) \cap \mathcal{E}(v_2).$$

An event is a triple $e = (t, r, o)$ of timestamp, issuer, and application operation. Applying a fresh event extends visibility from every event at the issuer head to e :

$$\text{vis}' = \text{vis} \cup (\mathcal{E}(H(r)) \times \{e\}).$$

Merge does not add an event and therefore leaves visibility unchanged.

Figure 2 gives all four constructors of **Step**. The notation $C[v \mapsto (S, E); r \mapsto v; P(v) \mapsto \pi]$ updates the version store, replica head, and parent list while preserving all other entries. The raw **APPLY** rule deliberately does not check whether the operation was issuable at that replica.

$$\begin{array}{c} \frac{H(r) = \perp}{C \xrightarrow{\text{create}(r)} C[r \mapsto 0]} \text{CREATE} \quad \frac{H(r) = v \quad V(v) = (S, E) \quad z = \text{query}(S, q)}{C \xrightarrow{\text{query}(r, q, z)} C} \text{QUERY} \\[10pt] \frac{H(r) = v \quad V(v) = (S, E) \quad t \notin \text{times}(C) \quad v' \notin \text{dom}(V) \quad v < v' \quad e = (t, r, o)}{C \xrightarrow{\text{apply}(t, r, o)} C[v' \mapsto (\text{do}(S, e), E \cup \{e\}); r \mapsto v'; P(v') \mapsto [v]]} \text{APPLY} \\[10pt] \frac{H(r_1) = v_1 \quad H(r_2) = v_2 \quad \text{LCA}_P(v_1, v_2) = v_T \quad V(v_i) = (S_i, E_i) \ (i \in \{1, 2, T\}) \quad v_m \notin \text{dom}(V) \quad v_1, v_2 < v_m}{C \xrightarrow{\text{merge}(r_1, r_2)} C[v_m \mapsto (\text{mergeL}(S_T, S_1, S_2), E_1 \cup E_2); r_1 \mapsto v_m; P(v_m) \mapsto [v_1, v_2]]} \text{MERGE} \end{array}$$

Figure 2: Paper-level raw operational semantics. The Lean rules also update the coherent replica caches. **CREATE** points a fresh replica at the shared initial version; **QUERY** is read-only.

The rules expose the separation that the rest of the paper uses. Raw execution says which version is read and which immutable version is created. It does not say which operations a client may mint or which datatype invariant should hold.

3 Why the earlier induction fails

The conflict relation induces an arbitration order lo . The first proof attempt used a relation computed from the entire configuration while linearizing a smaller version event set. That is unsound when a third event, outside the smaller set, absorbs an arbitration edge.

Write $e_1 \parallel_C e_2$ when neither event sees the other, and write $e_1 \text{ rc } e_2$ when the historical proof's conflict resolver chooses e_1 before e_2 . The repaired relation is indexed by the event set being linearized:

$$e_1 \text{ lo}_E e_2 \iff (e_1 \text{ vis } e_2 \wedge \neg(e_1 \rightleftharpoons e_2)) \vee (e_1 \parallel_C e_2 \wedge e_1 \text{ rc } e_2 \wedge \neg \exists e_3 \in E. e_2 \text{ vis } e_3 \wedge \neg(e_2 \rightleftharpoons e_3)).$$

Here $e_1 \rightleftharpoons e_2$ means that their updates commute. The final existential is the important repair: only an absorber inside E may erase an arbitration edge while E is being linearized.

Proposition 3.1 (global-order convergence is false). *There is a two-operation add-wins signature and a legal configuration with a backward-closed event set E such that two enumerations of E both respect the configuration-global arbitration relation but fold to different states.*
`convergence_over_backward_closed_subsets_false`

The checked countermodel is in `Metatheory/Join/Convergence_CounterModel.lean`. The repaired relation $\text{loOn } C \ E$ computes arbitration relative to E . In the countermodel it retains the edge that the outside absorber erased globally.

Associativity, commutativity, and idempotence of merge do not repair the proof. The second checked countermodel satisfies the core update laws and a bounded join-semilattice merge but violates both the peel laws and the Join lemma. `coreVCs_lattice_insufficient`

These countermodels establish two boundaries: arbitration must be local to the set being linearized, and convergence needs a contextual bridge between merge and update history.

Figure 3 summarizes the repair. The failed path tried to peel operations from a merge using a global order. The checked path instead proves canonicity for each stored event set and uses one contextual Join obligation at every merge.

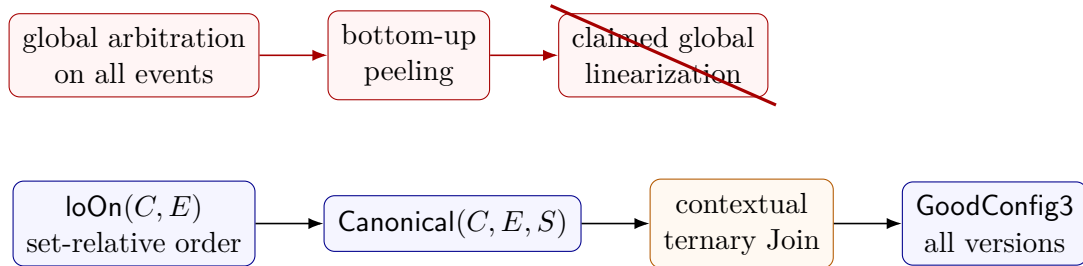


Figure 3: The failed bottom-up argument and the checked canonical-adequacy argument. Red nodes describe the refuted route; blue/orange nodes are current Lean interfaces and invariants.

4 Canonical adequacy

4.1 Canonical states

For a list $\pi = [e_1, \dots, e_n]$, let $\text{perm}(\pi, E)$ mean that π enumerates exactly E without duplicates. Let $\text{respects}(\pi, R)$ mean that no later list element is required by R to precede an earlier element.

Canonicity is then the following state predicate:

$$\text{Canonical}(C, E, s) \iff \exists \pi. \text{perm}(\pi, E) \wedge \text{respects}(\pi, \text{lo}_E) \wedge \text{fold}(\sigma_0, \pi) = s. \quad (\text{Canonical})$$

This is `IsCanonicalState`. The update-layer lemmas prove that any two such enumerations have the same fold under the supplied update conditions.

The reachable invariant `GoodConfig3` states that every registered version contains a canonical state of its event set. It also retains visibility transitivity and irreflexivity, event support, and causal closure. The invariant ranges over all versions, not only replica heads, because later merges may read an older version as their LCA.

The public correctness target uses the original configuration-wide order lo_C . The set-relative relation is stronger on E , so a canonical witness also satisfies the public target:

$$\text{RALin}(C) \iff \forall v, s, E. V(v) = (s, E) \Rightarrow \exists \pi. \text{perm}(\pi, E) \wedge \text{respects}(\pi, \text{lo}_C) \wedge \text{fold}(\sigma_0, \pi) = s. \quad (\text{RA-Lin})$$

Thus the strengthened invariant is internal to the corrected proof; the observable theorem remains per-version RA-linearizability.

4.2 The ternary Join obligation

Definition 4.1 (contextual ternary Join). *For a configuration C , suppose E_1 and E_2 satisfy the required closure conditions. If s_0 , s_1 , and s_2 are canonical for $E_1 \cap E_2$, E_1 , and E_2 , respectively, then*

$$\text{mergeL}(s_0, s_1, s_2)$$

must be canonical for $E_1 \cup E_2$.

JoinLemma3At

`JoinLemma3` quantifies this obligation over all configurations. Some datatypes prove the weaker per-configuration form from a history fact. The full-closure sibling `JoinLemma3F` supports algorithms whose merge proof needs complete causal closure rather than only non-commuting predecessors.

Theorem 4.2 (ordinary adequacy). *If a datatype satisfies `JoinLemma3`, every configuration reachable from `initConfig` by `Step` is RA-linearizable at every registered version.*
ra_linearizable3_of_join

The proof carries `StoreInv` with `GoodConfig3`. The store invariant proves that parent reachability remains well founded and that the selected LCA contains exactly the event-set intersection (`lca_events_of_storeInv`). `Apply` extends a canonical fold by a fresh event. `Merge` invokes the Join obligation. Query and replica creation preserve the invariant directly.

The framework retains the eight-condition route as one constructor for Join: `CoreVCs3CD`, `FeasibleDeltaVCs3`, and `CDVC3` imply `JoinLemma3`. Datatypes may instead prove Join directly; the public `VerifiedMRDT` package depends on the convergence result, not on one particular VC decomposition.

4.3 A checkable VC route to Join

The contextual Join lemma is the public convergence obligation. The framework also provides a reusable decomposition for datatypes whose ternary merge acts like applying deltas relative to an LCA. All equations below are quantified over canonical states for causally closed event sets; this restriction is load-bearing.

The first bundle contains the ordinary update laws and branch symmetry:

$$\text{mergeL}(l, a, b) = \text{mergeL}(l, b, a). \quad (\text{Core})$$

The feasible-delta bundle contains a unit law and two redistribution laws. A local event e appears only on branch 1:

$$\text{mergeL}(s_0, \text{mergeL}(B, t_1, B+e), s_2) = \text{mergeL}(B, \text{mergeL}(s_0, t_1, s_2), B+e). \quad (\text{Local redistribution})$$

For a shared event, the same delta appears in the LCA and both branches:

$$\begin{aligned} &\text{mergeL}(\text{mergeL}(B, t_0, B+e), \text{mergeL}(B, t_1, B+e), \text{mergeL}(B, t_2, B+e)) \\ &= \text{mergeL}(B, \text{mergeL}(t_0, t_1, t_2), B+e). \end{aligned} \quad (\text{Shared redistribution})$$

Here $B+e$ abbreviates $\text{do}(B, e)$. The canonical-delta equation peels a union-maximal event after merging its honest causal base:

$$\text{mergeL}(B, A, B+e) = A+e. \quad (\text{CD})$$

The premises identify B with the canonical state of $\downarrow e \setminus \{e\}$ and A with the canonical state of $U \setminus \{e\}$. They also require e to be maximal under lo_U .

Theorem 4.3 (VC adequacy). *CoreVCs3CD, FeasibleDeltaVCs3, and CDVC3 imply JoinLemma3, which in turn implies per-version RA-linearizability of every reachable raw configuration. join_lemma3_of_cd_feasible, ra_linearizable3_of_join*

This route is sufficient, not mandatory. A datatype may supply a direct Join proof, or use JoinLemma3F when its proof needs full causal closure.

5 Issuance, sequential meaning, and safety

RGA insertion illustrates why a plain signature still needs a client protocol. An insertion carries an anchor and a fresh identifier. Raw delivery can apply that record anywhere, but honest minting requires the issuer to have seen the anchor and to choose an identifier above its causal past. This is a historical fact about the minting state, not a shape predicate on an arbitrary later delivery state.

Definition 5.1 (issuance). *An Issuance D contains only*

$$\text{CanIssue} : \text{Op} \rightarrow \Sigma \rightarrow \text{Prop}.$$

IssuedStep checks **CanIssue** only for an apply transition and checks it at the issuer's current materialized head. Non-apply transitions retain their raw semantics. A final configuration does not store all former issuer heads, so **MintCertifiedReach** explicitly carries a causal-fold issuance witness at each step. **MintCertifiedReachV** is the virtual-LCA counterpart. A datatype derives any additional history invariant locally from that witness; the public interface does not store an arbitrary history field.

$$\begin{array}{c}
\frac{C \xrightarrow{\text{apply}(t,r,o)} C' \quad H(r) = v \quad V(v) = (S, E) \quad I.\text{CanIssue}((t, r, o), S)}{C \xrightarrow[\text{I}]{\text{apply}(t,r,o)} C'} \text{CERTIFIED-APPLY} \\
\\
\frac{C \xrightarrow{\ell} C' \quad \ell \neq \text{apply}(-, -, -)}{C \xrightarrow[\text{I}]{\ell} C'} \text{CERTIFIED-OTHER}
\end{array}$$

Figure 4: Issuance-certified execution. Certification restricts minting at the issuer head; it does not alter raw delivery or merge semantics. A `MintCertifiedReach` trace additionally retains the historical mint witness needed after the issuer moves to a later head.

Definition 5.2 (sequential specification). A *SequentialSpec* D supplies an abstract state, initial state, deterministic step, query function, and

$$\text{Legal} : \text{List}(\text{Op}) \rightarrow \text{Prop}.$$

The legality predicate sees only the abstract event history. It cannot inspect the implementation state. A total datatype sets it to true.

Definition 5.3 (semantic interaction). An *InteractionSpec* D classifies each ordered event pair as independent or as a conflict. A conflict carries one concurrent verdict: first-before-second, second-before-first, or unconstrained. Reversing the event pair must flip the verdict. This policy refers to abstract events, not arbitrary representation states.

For a version event set E , `interactionLoOn` orders every visible conflict by visibility. It orders a concurrent conflict only when the policy selects first-before-second and the second event has no later conflicting absorber inside E . This absorber clause belongs to the public witness order; it does not expose the proof-local replay resolver.

Two finite checks explain why this separation matters. LWW orders concurrent writes by increasing Lamport timestamp. Three writes therefore form a two-edge chain, which refutes the historical `no_rc_chain` requirement (`InteractionSPOT.LWW.old_no_chain_refuted`). For add-wins observed-remove semantics, concurrent remove and add conflict and the policy orders remove before add. If the remove observed the add, visibility instead orders add before remove. The same policy therefore explains both add-wins concurrency and ordinary observed removal.

Definition 5.4 (verified MRDT). An implementer supplies:

- an *Issuance*;
- an *InteractionSpec* that states semantic operation conflicts and any required concurrent direction;
- a *ConvergenceCertificate* for virtual-LCA certified execution, from which the ordinary theorem is derived; and
- an independent *SequentialSpec*, representation relation, and one *SequentialCorrectnessCertificate*.

VerifiedMRDT

Definition 5.5 (production package). A *PackagedMRDT* is a dependent pair (D, V) where D is one raw *MRDTSig* and V has type *VerifiedMRDT* D . The typed *Production.registry* contains only these pairs. A raw signature, replay-only certificate, or theorem about a different signature cannot enter the registry.

For every stored version (S, E) , both capstones produce a list π such that

$$\text{perm}(\pi, E) \wedge \text{respects}(\pi, \text{interactionLoOn}) \wedge \text{Legal}(\pi) \wedge \text{Rel}(S, \text{run}(\pi))$$

and every implementation query equals the sequential query. The package exposes `VerifiedMRDT.converges` and `VerifiedMRDT.convergesV`. The replay-only package remains available as `ReplayVerifiedMRDT` for internal replay proofs and checked negative classifications, but it is not the public sequential-correctness result. A datatype may separately supply a `SafetyCertificate`. It stores widened preservation; ordinary preservation follows from the ordinary-to-widened execution embedding. Safety remains separate because convergence does not imply a client invariant. For example, the bounded counter proves non-negative observable values through a global history invariant; arbitrary ternary tuples do not preserve that invariant.

The executable `CRDTSig` contains no resolver. The historical absorber-based convergence construction receives a proof-local `ReplayPolicy`. The generic certified Join route fixes its low-priority, unconstrained default; `IsRALinearizableWith` remains an internal hook for studying another replay proof. Consequently, neither `rc` nor `no_rc_chain` is an implementer-facing MRDT field.

6 Canonical virtual LCAs

Criss-cross histories can have several maximal common ancestors and no unique registered LCA. Choosing one maximal ancestor can lose events represented by another. The framework instead constructs a synthetic state by recursively folding the maximal-common-ancestor frontier.

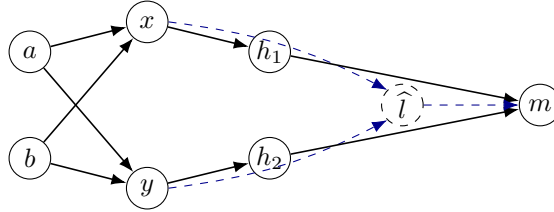


Figure 5: The registered DAG has two maximal common ancestors, x and y . The resolver constructs $S_{\hat{l}} = \text{foldMCA}(\{x, y\})$ for event set $\mathcal{E}(x) \cup \mathcal{E}(y)$; only merge result m is allocated. Solid arrows are stored parent edges; blue dashed arrows describe the ghost computation.

`canonicalVirtualLCA` implements this resolver. The support decreases under the version order, which justifies the recursion. The key theorem `virtualLCASState_canonical` proves that the synthetic state is canonical for the union of the maximal ancestors' event sets. The store theorem `mca_events_cover` shows that this union covers every common ancestor.

`StepV` conservatively embeds every ordinary `Step` and adds one virtual merge constructor. It allocates only the resulting version; the synthetic base is ghost semantic state.

Theorem 6.1 (virtual-LCA adequacy). *The same `JoinLemma3` that proves ordinary adequacy proves *RA-linearizability* for configurations reachable under `StepV` with the canonical resolver.*
`ra_linearizable3V_of_join`

No new datatype merge law is introduced by virtual LCAs.

7 Distributed commit-history garbage collection

Commit-history GC belongs to the runtime framework. It is not supplied once per datatype, and the public framework has no global stop-the-world collector.

7.1 Minimal local state

For each replica, the logical collector stores only

$$L = \{\text{head} : \text{Version}, \quad \text{commits} : \mathcal{P}(\text{Version})\}, \quad \text{head} \in \text{commits}.$$

The parent relation, fixed roster, and immutable optional author of each operation commit are protocol parameters. Roots and merge commits are unauthored. An envelope contains only a commit set. Fetch is set union and does not change the receiving head; head synchronization is a separate transition. **GC.Local**, **GC.Envelope**

A distributed world is only a replica-indexed family of these local records:

$$W : \text{Replica} \rightarrow \text{Local}.$$

In particular, the logical protocol does not store a second frontier map, an authorship index, or a copy of the global configuration at each replica. Efficient implementations may cache indexes, but the proof does not require them as semantic state.

Evidence is derived rather than stored. Replica r has frontier evidence at L when L retains a commit authored by r that reaches $L.\text{head}$. Formally,

$$\text{evidence}(L, r) = \{v \in L.\text{commits} \mid \text{author}(v) = r \wedge v \preceq L.\text{head}\}.$$

GC.EvidenceComplete requires such evidence for every other roster member. The negative control **GC.EvidenceSPOT.missing_author** confirms that an unauthored merge head cannot stand in for a missing author's evidence.

7.2 Collection certificate

A **GC.Certificate** supplies a keep set containing the local head and the evidence witnesses. It proves support in the local holding and exact reachability and LCA answers between retained versions. The canonical constructor **GC.Certificate.ofMCAClosed** uses a keep set closed under maximal common ancestors.

For local record L and keep set K , the certificate obligations are:

$$\begin{aligned} &L.\text{head} \in K, & K \subseteq L.\text{commits}, \\ &\forall r \in \text{roster}. r = \text{self} \vee \exists v \in K. v \in \text{evidence}(L, r), \\ &a, b \in K \Rightarrow (a \preceq_K b \iff a \preceq b), \\ &v_1, v_2, v_T \in K \Rightarrow (\text{LCA}_K(v_1, v_2, v_T) \iff \text{LCA}(v_1, v_2, v_T)). \end{aligned}$$

The compressed graph connects two retained versions when the original graph reached between them. It need not retain intermediate paths or the old root. **GC.compressedReaches_iff** proves reachability equivalence, and **GC.root_absent_when_dropped** proves that a root outside the keep set is absent from the retained carrier.

7.3 Direct asynchronous refinement

The collecting protocol has fetch, head synchronization, and local GC. Its no-GC counterpart has the same network steps and no collection transition. The simulation relates one full and one compact local store by equal heads and commit-set inclusion:

$$\text{StoreSim}(F, C) \equiv F.\text{head} = C.\text{head} \wedge C.\text{commits} \subseteq F.\text{commits} \wedge C.\text{head} \in C.\text{commits}.$$

$$\begin{array}{c} \frac{M = \text{commits}(W(s))}{W \longrightarrow W[d \mapsto \langle \text{head}(W(d)), \text{commits}(W(d)) \cup M \rangle]} \text{FETCH} \\ \frac{v \in \text{commits}(W(r))}{W \longrightarrow W[r \mapsto \langle v, \text{commits}(W(r)) \rangle]} \text{HEAD-SYNC} \quad \frac{\text{Certificate}(W(r), K)}{W \longrightarrow W[r \mapsto \langle \text{head}(W(r)), K \rangle]} \text{COLLECT} \end{array}$$

Figure 6: Asynchronous commit-history GC. Fetch is the ordinary remote-fetch path and unions immutable commits. Head synchronization is explicit. Collect is local and requires complete frontier evidence plus reachability/LCA preservation for the keep set.

Let NoGCStep contain only FETCH and HEAD-SYNC. The one-step simulation is

$$\text{WorldSim}(F, C) \wedge C \longrightarrow_{\text{GC}} C' \implies \exists F'. F \xrightarrow{\text{NoGC}}^{0 \text{ or } 1} F' \wedge \text{WorldSim}(F', C').$$

The collection case chooses $F' = F$; network cases take their matching no-GC step. Induction over the finite collecting trace gives the execution theorem.

Theorem 7.1 (distributed collection refines no-GC). *Every finite collecting execution is matched by a finite no-GC execution. Fetch and head synchronization match their counterparts; collection stutters. The final worlds have equal replica heads, so head-state reads are unchanged.*
 $\text{GC.execution_refines_noGC}, \text{GC.read_preserved}$

The runtime lifting adds a ghost MRDT configuration to state the semantic simulation. Visible apply and merge labels evolve that configuration. Fetch and GC remain silent. Erasing silent labels yields an ordinary or virtual-LCA core execution ($\text{GC.runtime_refines_core}, \text{GC.runtime_refines_coreV}$). The ghost configuration is not a second physical copy required of an implementation.

8 Optional datatype-state garbage collection

Commit GC removes versions from a replica's history. It cannot decide whether an RGA tombstone, a Peritext mark boundary, or a TreeMove event can be removed from the state stored at a retained version. That decision is datatype specific.

Definition 8.1 (state-GC protocol). *A StateGCProtocol $D \ V$ supplies*

$$\begin{array}{ll} \text{Physical} & : \text{Type}, \\ \text{semantic} & : \text{Physical} \rightarrow \text{Configuration}(D), \\ \text{Valid} & : \text{Physical} \rightarrow \text{Prop}, \\ \text{PhysicalStep} & : \text{Physical} \rightarrow \text{Option}(\text{Label}) \rightarrow \text{Physical} \rightarrow \text{Prop}. \end{array}$$

It proves:

1. *validity is invariant;*
2. *a silent physical transition leaves the semantic projection unchanged;*
3. *a visible transition refines a **StepV** transition.*

Theorem 8.2 (state collection refines semantic execution). *Erasing silent labels from any valid finite physical trace yields a finite **StepV** trace between the projected configurations. `StateGCProtocol.refines`*

The combined physical state contains the datatype protocol state and the generic distributed commit stores. A combined transition is either a datatype physical step or a silent fetch/commit-GC step. Writing $P = \langle p, W \rangle$, the two cases are

$$\frac{p \xrightarrow{\lambda}_S p' \quad W \xrightarrow{\lambda}_{\text{store}} W'}{\langle p, W \rangle \xrightarrow{\lambda} \langle p', W' \rangle} \text{ DATATYPE} \quad \frac{W \xrightarrow{\tau}_{\text{fetch/GC}} W'}{\langle p, W \rangle \xrightarrow{\tau} \langle p, W' \rangle} \text{ HISTORY.}$$

For $\lambda = \tau$, a datatype step leaves the commit store unchanged. For a visible label, its store evolution installs the version created by the semantic step. Both history transitions are silent at the datatype semantics.

Theorem 8.3 (the two collectors compose). *For every valid finite combined trace, erasing state-GC, fetch, and commit-GC labels yields an execution of the uncollected virtual-LCA semantics. `GC.CombinedSteps.refinesV`*

The ordinary corollary `GC.CombinedSteps.refinesRaw` applies when every visible datatype transition proves a raw **Step**. Sided Peritext and TreeMove provide concrete state-GC protocols. Other datatypes may omit the interface entirely.

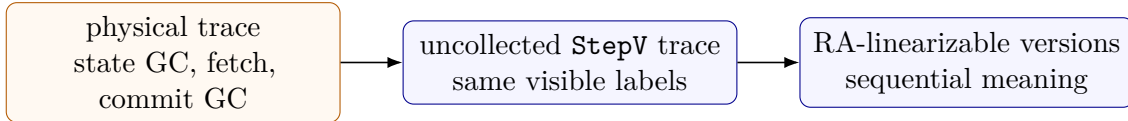


Figure 7: The two collectors remove different objects and compose by trace erasure. Commit GC removes versions; datatype GC removes metadata inside a retained state. The first arrow is `CombinedSteps.refinesV`; the second uses the `VerifiedMRDT` capstones.

9 Machine-checked instances

Table 1 lists the packages imported by `Metatheory/RefactorLedger.lean`. A checkmark means that the current Lean tree contains the named public package; it does not claim that the JavaScript runtime is extracted from that package.

The separate negative ledger checks the MVR single-register counterexample, the queue FIFO counterexample, and internal policy models. In particular, FugueMax’s coordinate-level `FMSig` has issuance, convergence, and RA-linearizability theorems, but it is not a public datatype package. Its role is to justify ordering-policy properties used by the released `ProductionRGA.sided` package. This distinction prevents a partial proof from appearing in the production table.

The production ledger also checks product construction, Peritext rendering, the distributed collector, and the combined GC theorem. The cited instance files print standard Lean axioms only; the repository release script rejects `sorry` and `sorryAx` in the production tree and validates the runtime evidence manifest.

Datatype	Issuance	Convergence	Public spec	State GC
Bounded counter	✓	✓	✓	–
RGA with tombstones	✓	✓	✓	–
EmbedRGA	✓	✓	✓	policy lemmas
SidedEmbedRGA	✓	✓	✓	policy lemmas
Observed-remove set	✓	✓	✓	–
Peritext/RichCore	✓	✓	✓	concrete protocol
TreeMove	✓	✓	✓	concrete protocol
AegisSheet	✓	✓	✓ ¹	certificate

Table 1: Current nontrivial production packages. ¹The AegisSheet theorem uses causal-origin legality because the old whole-prefix legality rejects valid concurrent origins; its reference state is causally aware, not only the visible sheet. Trivial-generation grow-only and counter products are also in the ledger.

AegisSheet illustrates why origin issuance and merged-history legality are separate. Two concurrent operations can each be valid at an empty origin, but neither was issued after observing the other. Rechecking the second operation against the full serialization prefix therefore rejects every order. The public `CausalOriginLegal` predicate instead requires an applicable causal origin contained in the prefix. The theorem `canonical_causalOriginLegal` constructs such a timestamp-ordered witness for every certified version. `spec_linearizable` and `spec_linearizableV` then prove that the independent incremental spreadsheet machine materializes the replicated state and returns the same queries under ordinary and virtual-LCA execution.

This incremental machine is not a conventional visible-sheet ADT. A checked abstraction audit constructs two reachable states with the same complete sheet view and applies the same honestly issued concurrent removal to both. The row disappears in one result and survives in the other because a same-position move created a hidden observed-remove keep token. Thus `no_view_only_step` rules out any deterministic transition function whose whole state is the visible view. Two companion witnesses perform the same test for a same-value cell overwrite and a same-range overwrite: the visible states agree before a concurrent selective overwrite, but their results differ because the active write identities differ. Consequently, observed-remove tokens and active cell/range version identities are semantic history required by the conflict and selective-undo policies, not event-log replay caches. The public theorem therefore claims refinement to a causally aware incremental spreadsheet machine, not to a visible-state single-user spreadsheet.

External implementation correspondence. We also audited the public Scala implementation of AegisSheet against the semantics published in its paper. At Bismuth commit `dd4c614`, four focused regressions fail. Two directly contradict Table 4 of the AegisSheet paper: undoing a move does not preserve the stated position of a concurrent insertion, and undoing one of two concurrent moves does not “revert both.” A column move can also cross a range without removing its live range identifier, after which the public `listRanges()` operation throws. Finally, undoing a range removal after a concurrent insertion restores stale numeric coordinates rather than the original stable endpoints. The untouched upstream suites pass 69 tests, and our merge-law and purge controls pass; the evidence therefore localizes these failures to the Scala undo and range layers rather than refuting the underlying merge kernel. The report and permanent reproducers are in `docs/aegissheet-scala-audit.md` and `audits/aegissheet/AegisSheetRegressionSuite.scala`. Because the published paper links

to a moving `main` branch rather than an immutable artifact revision, we must confirm the counterexamples and the relevant publication snapshot with the AegisSheet authors.

10 Reproduction and residual boundary

From the repository root, run the production proof and runtime gate with `./scripts/check-mrdt-refactor.sh`. Build both papers and the declaration ledger with `./scripts/check-working-papers.sh`.

The proved framework covers semantic MRDT executions, canonical virtual LCAs, distributed commit-history collection, optional state-GC protocols, and their composition. Same-replica crash recovery is not yet part of the verified or tested continuation protocol. The JavaScript implementation uses indexes and cached metadata for efficiency; the paper-facing logical GC state remains the minimal `{head, commits}` record.